Available online at www.sciencedirect.com
ScienceDirect
Theoretical Computer Science 364 (2006) 98–114
Theoretical Computer Science
www.elsevier.com/locate/tcs

Learning of erasing primitive formal systems from positive examples

Jin Uemura*, Masako Sato

Department of Mathematics and Information Sciences, Osaka Prefecture University, Sakai, Osaka 599-8531, Japan

Abstract

The present paper considers the learning problem of *erasing primitive formal systems*, PFSs for short, in view of inductive inference in Gold framework from positive examples. A PFS is a kind of logic program over strings called *regular patterns*, and consists of exactly two axioms of the forms $p(\pi) \leftarrow$ and $p(\tau) \leftarrow p(x_1), \ldots, p(x_n)$, where p is a unary predicate symbol, π and τ are regular patterns, and x_is are distinct variables. A PFS is *erasing* or *nonerasing* according to allowing the empty string substitution for some variables or not. We investigate the learnability of the class $\mathscr{PFSL}$ of languages generated by the *erasing* PFSs satisfying a certain condition. We first show that the class $\mathscr{PFSL}$ has M-finite thickness. Moriyama and Sato showed that a language class with M-finite thickness is learnable if and only if there is a finite tell tale set for each language in the class. We then introduce a particular type of finite set of strings for each *erasing* PFS, and show that the set is a finite tell tale set of the language. These imply that the class $\mathscr{PFSL}$ is learnable from positive examples.
© 2006 Published by Elsevier B.V.

Keywords: Inductive inference; Positive example; Erasing pattern language; Elementary formal system; Primitive formal system

1. Introduction

An *elementary formal system*, *EFS* for short, is a kind of logic program over *patterns* consisting of finitely many axioms. A pattern is a finite string consisting of constant symbols and variables. In EFSs, patterns are used for terms in first-order logic. For example, $\Gamma = \{p(ab) \leftarrow,\ p(axb) \leftarrow p(x)\}$ is an EFS with two axioms, where p is a unary predicate symbol, a and b are constant symbols, and x is a variable. In this example, the patterns ab and axb are used as terms. An EFS generates its language by finitely many applications of substitutions for the variables and modus ponens starting from the axioms in the EFS. In the above example, the language generated by Γ is $L(\Gamma) = \{a^n b^n \mid n \geqslant 1\}$. An EFS is *erasing* or *nonerasing* according to allowing the empty string substitution for some variables or not.

The framework of EFSs was originally introduced by Smullyan [18] to develop his recursive function theory, where no empty string is allowed, and thus they were nonerasing. This paper deals with erasing EFSs. Concerning with nonerasing EFSs, Arikawa et al. [4] introduced some subclasses of nonerasing EFSs whose language classes correspond to Chomsky hierarchy. Especially, the class of the language generated by a nonerasing EFS with just one axiom of the form $p(\pi) \leftarrow$ is called nonerasing *pattern language*. The class of nonerasing pattern language was first

* Corresponding author.
E-mail addresses: jin@mi.cias.osakafu-u.ac.jp (J. Uemura), sato@mi.cias.osakafu-u.ac.jp (M. Sato).

0304-3975/$ - see front matter © 2006 Published by Elsevier B.V.
doi:10.1016/j.tcs.2006.07.043

introduced by Angluin [2] as learnable class from positive examples in Gold framework [5]. Furthermore, the class of the nonerasing *length-bounded* EFSs that generates the class of the context-sensitive languages has been intensively investigated from the viewpoint of learning from positive examples [17,6,8,12]. Shinohara [15–17] showed that the class of the languages generated by the nonerasing length-bounded EFSs with at most n axioms has *finite elasticity*. Finite elasticity was introduced by Wright [20] as a sufficient condition for learnability from positive examples, and is closed under various class operations such as union, concatenation and so on [7,6,12].

On the other hand, concerning with *erasing* EFSs, even the learnability problem for the class of the *erasing* pattern languages was negatively solved if the number of the constant symbols is 2, 3 or 4 [10,11], and is open still now otherwise. The class of the *erasing regular* pattern languages, however, was positively solved by Shinohara [13], where a pattern is *regular* if each variable appears at most once in the pattern. As another positive result, the class of unions of at most k erasing *regular* pattern languages is shown to be efficiently learnable from positive examples by the present authors [19], Arimura et al. [3].

In this paper, we deal with the *erasing* EFSs, called *primitive formal systems* or PFSs for short, consisting of exactly two axioms of the forms $p(\pi) \leftarrow$ (the *base step*) and $p(\tau) \leftarrow p(x_1), \ldots, p(x_n)$ (the *induction step*), where π and τ are regular patterns, and $x_1, \ldots, x_n$ are all of the mutually distinct variables appearing in τ. When the regular pattern π in the base step is a constant string, we assume that there is no pair (i, u) of $i \geqslant 1$ and nonempty string u satisfying $\tau_\pi = (\pi u)^i \pi$, where τ_π is a string obtained from τ by substituting π to each variable appearing in τ.

The purpose of this paper is to show that the class $\mathcal{PFSL}$ of languages generated by such erasing PFSs is learnable from positive examples.

The class of *nonerasing* PFS languages was first introduced by Shinohara [14], and contains properly that of *nonerasing* regular pattern languages. If the class $\mathcal{PFSL}$ has finite elasticity as the class of *nonerasing* PFS languages does [17], the class $\mathcal{PFSL}$ is learnable from positive examples. In Section 3, the class $\mathcal{PFSL}$ is shown to have M-finite thickness, but not to have finite elasticity. The notion of *M-finite thickness* was introduced by Moriyama and Sato [6], and it is shown that a class with M-finite thickness is learnable from positive examples if and only if for each language in the class, there is a *finite tell tale* set of the language. By this result, it is enough for our purpose to show the existence of a finite tell tale set of each PFS language.

For an erasing PFS Γ, the language is given by $L(\Gamma) = L(\pi) \cup L'$, where $L(\pi)$ is the erasing pattern language generated by using only the base step $p(\pi) \leftarrow$, and L' is the set of strings obtained by at least once applying the induction step of Γ. In order to show the existence of a finite tell tale set of $L(\Gamma)$, in Section 4, we investigate various properties of the erasing PFS languages. An erasing PFS Γ is *reduced* if $L(\pi) \subsetneq L(\Gamma)$, that is, $L' \not\subseteq L(\pi)$. We first prove that Γ is reduced if and only if $L(\pi) \cap L' = \emptyset$. We also give another characterization theorem for the reduced PFSs. By the theorem, it can be shown that the decision problem of whether a given PFS is reduced or not is efficiently computable. We then consider the inclusion problem $L(\Gamma') \subseteq L(\Gamma)$ for given erasing PFSs Γ' and Γ, and solve it for some cases which play an important role in the proof of the existences of finite tell tale sets.

Section 5 introduces a particular finite subset for each PFS language. In terms of the above results, we prove that this set is a finite tell tale set of the language. These imply our main theorem in the present paper that the class $\mathcal{PFSL}$ is learnable from positive examples.

2. Erasing PFS languages

In this section, we first give the basic definitions and notations on the erasing PFS languages, and then show that an erasing PFS language can be expressed as a union of the erasing regular pattern languages associated with two erasing regular patterns contained in heads of the PFS.

Let Σ be a finite *alphabet*, X be a countable set of *variables*, and Π be a set of *predicate* symbols. We assume these sets Σ, X, Π are mutually disjoint. Each predicate symbol is associated with a positive integer called an *arity*. A *pattern* is a string (possibly the *empty* string ε) on $\Sigma \cup X$. An *atom* is an expression of the form $p(\pi_1, \ldots, \pi_n)$, where p is a predicate symbol with an arity n and patterns $\pi_1, \ldots, \pi_n$. A *definite clause* is a clause of the form $A \leftarrow B_1, \ldots, B_m$ $(m \geqslant 0)$, where $A, B_1, \ldots, B_m$ are atoms. The atom A is called the *head* and the part $B_1, \ldots, B_m$ the *body* of thedefinite clause.

Definition 1. An EFS, is a finite set of definite clauses. For an EFS Γ, each definite clause in Γ is called the *axiom* of Γ.

A *substitution* is a homomorphism from the patterns to the patterns that maps each symbol $a \in \Sigma$ to itself. We permit an ε substitution that can map some variables to the empty string ε. By $\pi\theta$, we denote the image of a pattern π by a substitution θ. For an atom $A = p(\pi_1, \ldots, \pi_n)$ and a clause $C = A \leftarrow B_1, \ldots, B_m$, we define $A\theta = p(\pi_1\theta, \ldots, \pi_n\theta)$ and $C\theta = A\theta \leftarrow B_1\theta, \ldots, B_m\theta$. A definite clause C is *provable from* an EFS Γ, denoted by $\Gamma \vdash C$, if C is obtained by finitely many (possibly 0) applications of substitutions and modus ponens. We define the language of an EFS Γ by $L(\Gamma, p) = \{w \in \Sigma^* \mid \Gamma \vdash p(w)\}$, where p is a unary predicate symbol.

Definition 2. An, EFS Γ is a *simple formal system*, an *SFS* for short, if each clause in Γ is of the form $p(\pi) \leftarrow q_1(x_1), \ldots, q_n(x_n)$, where p, q_is are unary predicate symbols and $x_1, \ldots, x_n$ are mutually distinct variables appearing in π. A pattern π is *regular* if each variable appears in π at most once. An SFS Γ is a *regular formal system*, an *RFS* for short, if all patterns in heads of clauses in Γ are regular. An RFS Γ is a *PFS*, if it contains exactly two clauses of the forms

$$p(\pi) \leftarrow \quad \text{and} \quad p(\tau) \leftarrow p(x_1), p(x_2), \ldots, p(x_n),$$

where $n \geqslant 1$ and $x_1, x_2, \ldots, x_n$ are all of the variables appearing in τ. The former is called the *base step* and the latter the *induction step* of Γ.

A language L is an erasing EFS (resp., SFS, RFS or PFS) language if $L = L(\Gamma, p)$ for some EFS (resp., SFS, RFS or PFS) Γ and some unary predicate symbol.

A language L is an *erasing regular pattern* language if $L = L(\Gamma, p)$ for some RFS $\Gamma = \{p(\pi) \leftarrow\}$.

It can be shown that the class of the erasing RFS languages corresponds to that of the context-free languages [4]. Thus, a PFS language is context-free, but not always regular. In fact, the following PFS language is not regular.

Example 3. Let us consider the PFS $\Gamma = \{p(\varepsilon) \leftarrow,\ p(axb) \leftarrow p(x)\}$, where $\Sigma = \{a, b\}$. Then as easily seen, $L(\Gamma, p) = \{a^n b^n \mid n \geqslant 0\}$.

In what follows, we omit the terms erasing and regular if no confusion occurs.In this paper, we deal with a class of PFS languages. In what follows, we fix a unary predicate symbol, say p, and denote $L(\Gamma, p)$ by $L(\Gamma)$ simply. Moreover, we denote a PFS $\Gamma = \{p(\pi) \leftarrow,\ p(\tau) \leftarrow p(x_1), \ldots, p(x_n)\}$ and $L(\Gamma)$, by $\Gamma = (\pi, \tau)$ and $L(\pi, \tau)$, respectively. Similarly by $L(\pi)$, we denote $L(\{p(\pi) \leftarrow\})$.

2.1. Expression of a PFS language by a union of pattern languages

Let $\Gamma = (\pi, \tau)$ be a PFS. By the definition of $L(\Gamma)$, it follows that $L(\Gamma) = L(\pi) \cup L'$, where L' is the set of strings obtained by at least once applying the induction step of Γ. We show that the language L' as an infinite union of regular pattern languages associated with two regular patterns π, τ contained in heads of the PFS: let $var(\tau) = \{x_1, \ldots, x_n\}$, where $var(\tau)$ be the set of the variables appearing in τ, and let

$$\tau_\pi = \tau\{x_1 := \pi, x_2 := \pi, \ldots, x_n := \pi\},$$

where variables substituted to the variables x_is in τ are taken to be distinct, and so τ_π is always regular. Then we define $\Gamma_\tau = \bigcup_{t=1}^{\infty} \Gamma_t$, where Γ_t is recursively defined as follows: $\Gamma_1 = \{\tau_\pi\}$ and for each $t \geqslant 2$,

$$\Gamma_t = \{\tau\{x_1 := \xi_1, x_2 := \xi_2, \ldots, x_n := \xi_n\} \mid \xi_i \in \Gamma_{t-1} \cup \{\pi\}, i = 1, 2, \ldots, n\}.$$

By the above definition, clearly $\Gamma_{t-1} \subseteq \Gamma_t$ holds for $t \geqslant 2$ and the equality is valid if and only if $\tau = x$. Moreover each ξ in Γ_τ always contains π as a substring because of $var(\tau) \neq \emptyset$. By the definitions of $L(\Gamma)$ and the above Γ_τ, it follows that:

Lemma 4. *Let $\Gamma = (\pi, \tau)$ be a PFS. Then $L(\Gamma) = L(\pi) \cup L(\Gamma_\tau)$ holds, where $L(\Gamma_\tau) = \bigcup_{\xi \in \Gamma_\tau} L(\xi)$.*

Proof. The proof of $L(\Gamma_t) \subseteq L(\Gamma)$ for $t \geqslant 1$ can be easily done by mathematical induction on $t \geqslant 1$. Thus we have $L(\Gamma_\tau) \subseteq L(\Gamma)$.

Conversely, let w be a string in $L(\Gamma)$. By the definition of $L(\Gamma)$, $p(w)$ is provable from Γ, that is, $\Gamma \vdash p(w)$. Let $k \geqslant 0$ be the number of applications of the induction step $p(\tau) \leftarrow p(x_1), \ldots, p(x_n)$ for $\Gamma \vdash p(w)$. Then we can easily prove $w \in L(\pi) \cup L(\Gamma_\tau)$ by mathematical induction on $k \geqslant 0$. □

2.2. Canonical patterns and reduced PFSs

For patterns π and τ, we introduce binary relations $\preccurlyeq$ and $\equiv$ as follows: $\pi \preccurlyeq \tau$ if $\pi = \tau\theta$ for some substitution θ, and $\pi \equiv \tau$ if $\pi \preccurlyeq \tau$ and $\tau \preccurlyeq \pi$. A *renaming* of variables is a substitution θ such that $x\theta \in X$ and $x \neq y$ implies $x\theta \neq y\theta$ for any $x, y \in X$. In this paper, we do not distinguish two patterns equivalent to each other by renaming. Thus $ax_1bx_2 = ax_2bx_1$ and $ax_1bx_2 \equiv ax_1x_2bx_3$ but $ax_1bx_2 \neq ax_1x_2bx_3$.

A pattern π is of *canonical form* if: (i) $\pi \equiv \tau$ implies $|\pi| \leqslant |\tau|$ for any pattern τ; (ii) π contains exactly n variables $x_1, \ldots, x_n$ for some integer n and (iii) the leftmost occurrence of x_i is to the left of the leftmost occurrence of x_{i+1} for each i, where $|\pi|$ is the length of the pattern π.

Lemma 5 (*Shinohara [13]*). *Suppose that* $\sharp\Sigma \geqslant 3$. *Let* π *and* τ *be regular patterns. Then*

$$\text{(i) } \pi \preccurlyeq \tau \iff L(\pi) \subseteq L(\tau), \quad \text{(ii) } \pi \equiv \tau \iff L(\pi) = L(\tau).$$

Concerning with a regular pattern language, Shinohara [13] showed that for each regular pattern π, there is the unique canonical pattern π' equivalent to π using the above result, provided that $\sharp\Sigma \geqslant 3$. Clearly the canonical pattern π' has the form of $w_0x_1w_1x_2 \ldots w_{n-1}x_nw_n$ where $w_0, w_n \in \Sigma^*$ and $w_i \in \Sigma^+$ $(i = 1, \ldots, n-1)$, and $L(\pi) = L(\pi')$ holds.

Hereafter, we assume that $\sharp\Sigma \geqslant 3$. Then for a PFS $\Gamma = (\pi, \tau)$, can we assume π and τ to be of canonical form? By Lemma 4, $L(\Gamma) = L(\pi) \cup (\bigcup_{\xi \in \Gamma_\tau} L(\xi))$ holds. Since π and each $\xi \in \Gamma_\tau$ are regular, we have $L(\Gamma) = L(\pi') \cup (\bigcup_{\xi \in \Gamma_\tau} L(\xi'))$, where π' and ξ' are the canonical patterns of π and ξ, respectively. It implies that $L(\pi, \tau) = L(\pi', \tau)$, that is, we can assume π to be of canonical form. On the other hand, concerning with the pattern τ in the induction step, we cannot assume τ to be of canonical form. Indeed, let $\Gamma = (aa, bx_1x_2b)$. As easily seen, $b(aa)(aa)b \in L(\Gamma)$. On the other hand, for $\Gamma' = (aa, bx_1b)$, we have $b(aa)(aa)b \notin L(\Gamma')$. Thus $L(\Gamma) \neq L(\Gamma')$. In Corollary 15, each pattern $\xi \in \Gamma_\tau$ is shown to be of canonical form for a *reduced* PFS Γ defined below.

Definition 6. A PFS $\Gamma = (\pi, \tau)$ is reduced if $L(\pi) \subsetneq L(\Gamma)$.

By the above definition, Lemmas 4 and 5(i), a PFS Γ is reduced if and only if $L(\xi) \not\subseteq L(\pi)$ for some $\xi \in \Gamma_\tau$ if and only if $\xi \not\preccurlyeq \pi$ for some $\xi \in \Gamma_\tau$. If $\tau = x$ then $L(\pi, x) = L(\pi)$ holds, and thus $\Gamma = (\pi, x)$ is not reduced. Conversely, if $\Gamma = (\pi, \tau)$ is not reduced then $L(\Gamma) = L(\pi, x)$. If $\pi \in \Sigma^*$ and $\tau \neq x$, since $\tau_\pi \notin L(\pi)$, the PFS $\Gamma = (\pi, \tau)$ is always reduced .

In Section 4, we present characterization theorems for a reduced PFS, and show that the decision problem of whether $\Gamma = (\pi, \tau)$ is reduced is computable in time $O(|\pi| + |\tau|)$.

Finally, we impose some syntax-restriction to $\Gamma = (\pi, \tau)$. For two strings $w, u \in \Sigma^*$, $w \preccurlyeq_p u$ denotes that w is a prefix of u, and $w \preccurlyeq_s u$ denotes that w is a suffix of u. Moreover, $w \preccurlyeq_p u^*$ denotes that $w \preccurlyeq_p u^i$ for some $i \geqslant 0$. Similarly, we define $w \preccurlyeq_s u^*$. Let w, u be nonempty strings with $w \preccurlyeq_p u^*$. Then there is a unique pair (i, u') such that $w = u^i u'$, $u' \preccurlyeq_p u$ and $u' \neq u$. By $\rho_p(w, u)$, we denote the string u' for $w \preccurlyeq_p u^*$. Similarly we define $\rho_s(w, u)$. A string $w \in \Sigma^+$ is a *multiple string of a string* u, called a *component*, if $w = u^l$ for some $l \geqslant 2$, and is *multiple* if there is a component of w.

We denote by $\mathcal{PFS}$ the set of all PFSs except for PFSs $\Gamma = (\pi, \tau)$ such that $\pi \in \Sigma^*$, and $\tau_\pi \preccurlyeq_p u^*$ and $\rho_p(\tau_\pi, u) = \pi$ for some nonmultiple string u with $|u| < |\tau_\pi|$, and by $\mathcal{PFSL}$ the class of the languages generated by the PFSs in $\mathcal{PFS}$. Note that $\Gamma = (\varepsilon, \tau) \in \mathcal{PFS}$ if and only if the string τ_ε is nonmultiple.

3. Inductive inference

In this section, we first give the notions of identification in the limit from positive examples [5,2], and finite elasticity due to Wright [20] and M-finite thickness due to Moriyama and Sato [6] closely related to the learnability. Then we show that the class $\mathcal{PFSL}$ has M-finite thickness but not finite elasticity.

A language class $\mathcal{L} = L_0, L_1, \ldots$ over Σ is an *indexed family of recursive languages* if there is a computable function $f : N \times \Sigma^* \to \{0, 1\}$ such that $f(i, w) = 1$ if $w \in L_i$, 0 otherwise, where $N = \{i \mid i \geqslant 0\}$. The function f is called a *membership* function. Hereafter, we confine ourselves to indexed families of recursive languages.

An infinite sequence of strings $w_1, w_2, \ldots$ over Σ is a *positive presentation* of a language L, if $L = \{w_n \mid n \geqslant 1\}$ holds. An *inference machine* is an effective procedure M that runs in stages $1, 2, \ldots$, and at each stage, it requests an *example* and produces a *hypothesis* in N based on the examples so far received. Let M be an inference machine and $\sigma = w_1, w_2, \ldots$ be an infinite sequence of strings. We denote by h_n the hypothesis produced by M at stage n after the examples $w_1, \ldots, w_n$ are fed to M. M on input σ *converges* to h if there is an integer $n_0 \in N$ such that $h_n = h$ for every $n \geqslant n_0$. M *identifies in the limit* or *infers* a language L *from positive examples*, if for any positive presentation σ of L, M on input σ converges to h with $L = L_h$. A class of languages $\mathcal{L}$ is *inferable from positive examples* if there is an inference machine that infers any language in $\mathcal{L}$ from positive examples.

Angluin [2] gave a characterizing theorem for a language class $\mathcal{L}$ to be inferable from positive examples. A set S_L is called a *finite tell tale set* of L within $\mathcal{L}$ if S_L is finite and $S_L \not\subseteq L'$ for all $L' \in \mathcal{L}$ with $L' \subsetneq L$. A language class $\mathcal{L}$ inferable from positive examples if and only if there exists an effective procedure that enumerates for every language in $\mathcal{L}$ a set S_L such that $S_L \subseteq L$.

Recently, Mukouchi has showed that the class of SFS languages defined by at most two clauses is not inferable from positive examples by considering the following infinite sequence of SFSs:

$$\Gamma_n = \{p(x_1x_2 \ldots x_{n-1}x_nx_nx_{n-1} \ldots x_2x_1) \leftarrow\} \quad (n \geqslant 1).$$

We should note that the pattern $x_1x_2 \ldots x_{n-1}x_nx_nx_{n-1} \ldots x_2x_1$ is not regular, and thus the above EFSs are not RFSs, but SFSs. He showed $L(\Gamma_1) \subsetneq L(\Gamma_2) \subsetneq \cdots \subsetneq L(\Gamma)$ and $\bigcup_{n=1}^{\infty} L(\Gamma_n) = L(\Gamma)$ under $\sharp\Sigma \geqslant 2$, where

$$\Gamma = \{p(x_1x_1) \leftarrow, p(x_1x_2x_1) \leftarrow p(x_2)\}.$$

It means that the language $L(\Gamma)$ does not have any finite tell tale set within the class.

Angluin [2] gave a very useful sufficient condition for inferability called *finite thickness*. The class of the erasing regular pattern languages discussed in this paper was also shown to have finite thickness by Shinohara [13]. Wright [20] introduced another sufficient condition for inferability called *finite elasticity* more general than finite thickness [7]. A class $\mathcal{L}$ has *finite elasticity*, if there is no infinite sequence of strings $w_0, w_1, \ldots$ and no infinite sequence of languages $L_1, L_2, \ldots$ in $\mathcal{L}$ satisfying $\{w_0, w_1, \ldots, w_{n-1}\} \subseteq L_n$ but $w_n \notin L_n$ for every $n \geqslant 1$. Finite elasticity has a good property in a sense that it is closed under various class operations such as union, intersection and so on [6,12,20]. Shinohara [17] proved that the class of the *nonerasing* length-bounded EFS languages generated by at most k clauses has finite elasticity, and so is inferable from positive examples.

Mukouchi recently has showed that the class of *erasing* RFS languages generated by at most k clauses has finite elasticity similarly, provided that all regular patterns in heads of the induction steps are of canonical forms. Without the condition of canonical form, the inferability, however, is not valid [9].

As well as the class of the RFS languages above, our class $\mathcal{PFSL}$ does not have finite elasticity as shown below.

Theorem 7. *The class $\mathcal{PFSL}$ does not have finite elasticity.*

Proof. Define PFSs $\Gamma_n = (\varepsilon, \tau_n)$ $(n \geqslant 1)$ as follows: $\tau_n = a(x_1x_2 \cdots x_n)b$ for $n \geqslant 1$, where $a, b \in \Sigma$, $a \neq b$. Then we can show that

$$\{\varepsilon, ab, a(ab)^2b, \ldots, a(ab)^kb\} \subseteq L(\Gamma_k) \text{ but } a(ab)^{k+1}b \notin L(\Gamma_k) \text{ for } k \geqslant 1.$$

Thus, the infinite sequence $(w_n)_{n \geqslant 0}$ of strings and the infinite sequence $(\Gamma_n)_{n \geqslant 1}$ of PFSs satisfies the above conditions, where $w_0 = \varepsilon$, $w_n = a(ab)^nb$ $(n \geqslant 1)$. Hence, the class $\mathcal{PFSL}$ does not have finite elasticity. □

Moriyama and Sato [6] introduced a notion of M-finite thickness which is generalized notion of finite thickness. For a language S, a language L is a *minimal* language of S within $\mathcal{L}$ if $S \subseteq L$, and if no language $L' \in \mathcal{L}$ satisfies $S \subseteq L' \subsetneq L$. For nonempty finite set $S \subseteq \Sigma^*$, we define

$$\mathrm{MIN}(S, \mathcal{L}) = \{L \in \mathcal{L} \mid L \text{ is a minimal language of } S \text{ within } \mathcal{L}\}.$$

Definition 8. A class $\mathcal{L}$ has *M-finite thickness*, if for any nonempty set $S \subseteq \Sigma^*$, (i) $\sharp\mathrm{MIN}(S, \mathcal{L}) < \infty$, (ii) $S \subseteq L \in \mathcal{L}$ implies that there is a language $L' \in \mathrm{MIN}(S, \mathcal{L})$ such that $L' \subseteq L$.

M-finite thickness by itself is not a sufficient condition for inferability from positive examples, but is closed under various class operations such as union, concatenation and so on as well as finite elasticity [6,12].

Theorem 9 (*Moriyama and Sato [6]*). *If a class $\mathcal{L}$ has M-finite thickness, then the class $\mathcal{L}$ is inferable from positive examples if and only if for each language $L \in \mathcal{L}$, there is a finite tell tale set of L.*

Hereafter, we prove that our class $\mathcal{PFSL}$ has M-finite thickness. By Lemma 4, $L(\pi, \tau) = L(\pi) \cup L(\Gamma_\tau)$ holds. Hence for a string $w \in \Sigma^*$, $w \in L(\Gamma)$ implies $w \in L(\pi)$ or $w \in L(\Gamma_\tau)$. The former gives $|c(\pi)| \leqslant |w|$, and there are at most finitely many such πs since π is of canonical form. On the other hand, the latter gives $|c(\tau_\pi)| \leqslant |w|$, and thus $|c(\tau_\pi)| = |c(\tau)| + n|c(\pi)| \leqslant |w|$, where $n \geqslant 1$ denotes the number of variables in τ. If $\pi \neq \varepsilon$, there are at most finitely many such PFSs.

Theorem 10. *The class $\mathcal{PFSL}$ has M-finite thickness.*

Proof. Let $S \subseteq \Sigma^*$ be a nonempty finite set. Let $l_{\max}$ be the length of the longest strings in S.

Claim A. $\sharp\mathrm{MIN}(S, \mathcal{PFSL}) < \infty$ *holds.*

The proof of Claim A. Let $L(\Gamma) \in \mathrm{MIN}(S, \mathcal{PFSL})$ for $\Gamma = (\pi, \tau)$. If $\pi \neq \varepsilon$, as mentioned above, there are at most finitely many such PFSs because of $|c(\pi)| \leqslant l_{\max}$ if $S \subseteq L(\pi)$, and $|c(\tau_\pi)| \leqslant l_{\max}$ otherwise.

We first consider a case of $\pi = \varepsilon$ and $l_{\max} \geqslant 1$. In this case, $c(\tau) = \tau_\pi$ holds. Similar to the above, we have $|c(\tau)| \leqslant l_{\max}$, and thus there are at most finitely many such constant strings $c(\tau)$. Let us put $l = |c(\tau)|$. Then as easily seen, the lengths of the strings in $L(\varepsilon, \tau)$ are multiples of l. Thus, $l_{\max} = kl$ for some $k \geqslant 1$.

Let us put $\tau = w_0 X_1 w_1 \ldots w_{n-1} X_n w_n$, where $w_0, w_n \in \Sigma^*$, $w_i \in \Sigma^+ (i = 1, \ldots, n-1)$ and $X_i \in X^+$ for $i = 1, \ldots, n$. A variable $x \in var(\tau)$ is *nonerasable* w.r.t. S if $S \not\subseteq L(\varepsilon, \tau\{x := \varepsilon\})$. Since $L(\Gamma)$ is a minimal language of S, we can assume that every variable in τ is nonerasable w.r.t. S, and such τ is called a nonerasable pattern w.r.t. S. Clearly, $|X_i| \leqslant k$ holds for every i. Hence, there are at most finitely many PFS languages in $\mathrm{MIN}(S, \mathcal{PFSL})$.

Claim B. *For any PFS Γ, if $S \subseteq L(\Gamma)$, then $L(\Gamma') \subseteq L(\Gamma)$ for some $\Gamma' \in \mathrm{MIN}(S, \mathcal{PFSL})$.*

The proof of Claim B. Let $S \subseteq L(\Gamma)$ for $\Gamma = (\pi, \tau)$. We can assume that Γ is reduced, and $L(\Gamma) \notin \mathrm{MIN}(S, \mathcal{PFSL})$. Then we have $S \subseteq L(\Gamma') \subsetneq L(\Gamma)$ for some reduced PFS $\Gamma' = (\alpha, \beta)$. Similarly to the proof of Claim A, it can be shown that there are at most finitely many such $L(\Gamma')$s containing the set S. □

By Theorems 9 and 10, if there is a finite tell tale set for every language in the class $\mathcal{PFSL}$, the class is inferable from positive examples. In what follows, we investigate finite tell tale sets for PFS languages.

4. Reduced PFSs and inclusion problem for PFS languages

In this section, we deal with a PFS $\Gamma = (\pi, \tau)$ with $\pi \notin \Sigma^*$. As mentioned in Section 2, if $\pi \in \Sigma^*$ and $\tau \neq x$, then the PFS $\Gamma = (\pi, \tau)$ is always reduced. We first investigate properties of reduced PFSs and show that a reduced PFS has a very characteristic feature. That is, a PFS $\Gamma = (\pi, \tau)$ is reduced if and only if $L(\pi) \cap L(\Gamma_\tau) = \emptyset$. It means that if $w \in L(\Gamma)$ but $w \notin L(\pi)$ for some w, then the PFS Γ is reduced, and thus $L(\pi) \cap L(\Gamma_\tau) = \emptyset$. In terms of this result, we then investigate the inclusion problem for PFS languages.

For a pattern $\pi \notin \Sigma^*$, A_π and B_π denote the longest constant prefix and the longest constant suffix of π, respectively. For instance, $A_\pi = ab$ and $B_\pi = \varepsilon$ if $\pi = abx_1aax_2$.

The strings A_ξ and B_ξ for a pattern $\xi \in \Gamma_\tau$ can be expressed in terms of A_π, A_τ, B_π and B_τ as follows:

Lemma 11. *Let $\Gamma = (\pi, \tau)$ be a PFS with $\pi \notin \Sigma^*$. For any $\xi \in \Gamma_\tau$, there is a pair (i_0, j_0) of positive integers such that*

$$A_\xi = A_\tau^{i_0} A_\pi, \quad B_\xi = B_\pi B_\tau^{j_0}.$$

Proof. Let $\xi \in \Gamma_\tau$. Then there is an integer $t \geqslant 1$ such that $\xi \in \Gamma_t$. The proof of our lemma can be done by mathematical induction on $t \geqslant 1$. For $t = 1$, since $\xi = \tau_\pi$, $A_\xi = A_\tau A_\pi$ and $B_\xi = B_\pi B_\tau$ holds. Let $t \geqslant 2$ and let $\xi = \tau\{x_1 := \xi_1, \ldots, x_n := \xi_n\}$ for some $\xi_k \in \Gamma_{t-1}$ $(k = 1, \ldots, n)$. By $\xi_k \in \Gamma_{t-1}$ and induction hypothesis, there are positive integers $i_k, j_k \geqslant 1$ such that $A_{\xi_k} = A_\tau^{i_k} A_\pi$ and $B_{\xi_k} = B_\pi B_\tau^{j_k}$. Hence, we have $A_\xi = A_\tau A_{\xi_1} = (A_\tau)^{i_1+1} A_\pi$ and $B_\xi = B_{\xi_1} B_\tau = B_\pi (B_\tau)^{j_1+1}$. □

4.1. Characterization theorems for reduced PFSs

In this paragraph, we give theorems characterizing a reduced PFS $\Gamma = (\pi, \tau)$ with $\pi \notin \Sigma^*$.

By Pref, we denote the set of pairs (v, w) of strings satisfying $v \preccurlyeq_p w$ or $w \preccurlyeq_p v$. Similarly we define the set Suff. Clearly $\xi \preccurlyeq \pi$ implies $A_\pi \preccurlyeq_p A_\xi$ and $B_\pi \preccurlyeq_s B_\xi$. Hence, by Lemma 11, $L(\xi) \subseteq L(\pi)$ for $\xi \in \Gamma_\tau$ implies $A_\pi \preccurlyeq_p A_\tau^{i_0} A_\pi (= A_\xi)$ and $B_\pi \preccurlyeq_s B_\pi B_\tau^{j_0} (= B_\xi)$ for some $i_0, j_0 \geqslant 1$. The next lemma for patterns yields us the converse.

Lemma 12. *Let $\pi = A_\pi(x\pi' x')B_\pi$ and $\xi = A_\xi(x\xi' x')B_\xi$ for some patterns π', ξ' (possibly ε). Then the followings are valid.*

(i) $L(\pi) \cap L(\xi) \neq \emptyset \iff (A_\pi, A_\xi) \in \text{Pref}$ *and* $(B_\pi, B_\xi) \in \text{Suff}$.

(ii) *If π' is a substring of ξ', then*

$$L(\xi) \subseteq L(\pi) \iff A_\pi \preccurlyeq_p A_\xi, \quad B_\pi \preccurlyeq_s B_\xi.$$

Proof. The proof of the assertion (i). The "only if" part is obvious. We prove the "if" part for a case of $|A_\pi| \leqslant |A_\xi|$ and $|B_\pi| \leqslant |B_\xi|$. The other cases can be proved similarly. Assume that $(A_\pi, A_\xi) \in \text{Pref}$ and $(B_\pi, B_\xi) \in \text{Suff}$. Then $A_\xi = A_\pi A$ and $B_\xi = B B_\pi$ hold for some strings A and B. Consider a regular pattern $\gamma = A_\pi A \xi' \pi' B B_\pi$. Since $\gamma = \pi\{x := A\xi', x' := B\} = \xi\{x := \varepsilon, x' := \pi'\}$, it follows that $\gamma \preccurlyeq \pi$ and $\gamma \preccurlyeq \xi$. By Lemma 5, we have $\emptyset \neq L(\gamma) \subseteq L(\pi) \cap L(\xi)$.

The proof of the assertion (ii) can be done similar to the above. □

Lemma 13. *For $w \in \Sigma^*$ and $v \in \Sigma^+$,*

(i) $\exists\, i_0 \geqslant 1$ *s.t.* $w \preccurlyeq_p v^{i_0} w \iff w \preccurlyeq_p v^* \iff \forall i \geqslant 0,\ w \preccurlyeq_p v^i w$,

(ii) $\exists\, j_0 \geqslant 1$ *s.t.* $w \preccurlyeq_s w v^{j_0} \iff w \preccurlyeq_s v^* \iff \forall j \geqslant 0,\ w \preccurlyeq_s w v^j$.

Proof. We prove only the assertion (i).

Suppose $w \preccurlyeq_p v^{i_0} w$ for some $i_0 \geqslant 1$. Let l be the least integer satisfying $|w| < |v^{i_0 \cdot (l+1)}|$, and let w' be the prefix of w with length $|v^{i_0 \cdot l}|$ and $w = w'w''$ for some w''. Let us put $w' = w_1 w_2 \ldots w_l$ for some w_i with length $|w_i| = |v^{i_0}| (i = 1, \ldots, l)$. Then by the assumption, we have $w_1 \ldots w_l w'' \preccurlyeq_p v^{i_0} w_1 \ldots w_l w''$. It means that $w_1 = v^{i_0}$, $w_2 = w_1, \ldots, w_l = w_{l-1}$ and $w'' \preccurlyeq_p w_l$. Thus we have $w_i = v^{i_0}$ for $i = 1, \ldots, l-1$ and $w'' \preccurlyeq_p v^{i_0}$. It implies that $w = v^{i_0 \cdot l} w'' \preccurlyeq_p v^{i_0 \cdot (l+1)}$, and so $w \preccurlyeq_p v^*$. The other implications are obvious. □

By Lemmas 11–13, we obtain the following characterization theorem for a reduced PFS.

Theorem 14. *Let $\Gamma = (\pi, \tau)$ be a PFS with $\pi \notin \Sigma^*$. Then the following statements are equivalent*:

(i) *Γ is reduced.*

(ii) *$A_\pi \not\preccurlyeq_p (A_\tau)^{i_0} A_\pi$ or $B_\pi \not\preccurlyeq_s B_\pi (B_\tau)^{i_0}$ for some $i_0 \geqslant 1$.*

(iii) $L(\pi) \cap L(\Gamma_\tau) = \emptyset$.

Proof. We assume $L(\xi) \subseteq L(\pi)$. Obviously we have $L(\xi) \cap L(\pi) \neq \emptyset$. By Lemmas 11 and 12(i), $A_\pi \preccurlyeq_p A_\tau^{i_0} A_\pi$ $(= A_\xi)$, $B_\pi \preccurlyeq_s B_\pi B_\tau^{j_0} (= B_\xi)$ is valid. We note that each pattern $\xi \in \Gamma_\tau$ contains the pattern π as a substring. Hence, by Lemma 12(ii), it follows that:

$$L(\xi) \cap L(\pi) \neq \emptyset \iff L(\xi) \subseteq L(\pi) \iff A_\pi \preccurlyeq_p A_\tau^{i_0} A_\pi (= A_\xi), \quad B_\pi \preccurlyeq_s B_\pi B_\tau^{j_0} (= B_\xi)$$

for some $i_0, j_0 \geqslant 1$. If Γ is reduced, there is a pattern $\xi \in \Gamma_\tau$ satisfying $L(\xi) \not\subseteq L(\pi)$. By the above, the assertion (ii) is valid.

(ii) $\Rightarrow$ (iii). Assume that $L(\pi) \cap L(\Gamma_\tau) \neq \emptyset$ and $L(\xi) \cap L(\pi) \neq \emptyset$ for some $\xi \in \Gamma_\tau$. By Lemma 13, the following holds:

(ii′) $A_\pi \not\preccurlyeq_p (A_\tau)^i A_\pi$ or $B_\pi \not\preccurlyeq_s B_\pi (B_\tau)^i$ for any $i \geqslant 0$.

Then by the above, it contradicts (ii′). Clearly (iii) implies (i). □

Corollary 15. *Let $\Gamma = (\pi, \tau)$ be a PFS with $\pi \notin \Sigma^*$.*
(i) *If $A_\tau = B_\tau = \varepsilon$ or $A_\pi = B_\pi = \varepsilon$, then Γ is not reduced.*
(ii) *If $\Gamma = (\pi, \tau)$ is reduced, then any pattern $\xi \in \Gamma_\tau$ does not contain successive variables.*

The assertion (ii) holds because of $A_\pi \neq \varepsilon$ or $B_\pi \neq \varepsilon$ for a reduced PFS. Remember that the regular pattern π for a PFS $\Gamma = (\pi, \tau)$ is assumed to be of canonical form. Thus, by the assertion, every pattern in Γ_τ is of *canonical* if Γ is reduced.

Let us put $w = A_\pi$, $v = A_\tau$ (resp. $w = B_\pi$, $v = B_\tau$). Then Lemmas 13, 11 and Theorem 14 yield us the following characterizing theorem for reduced PFSs.

Theorem 16. *Let $\Gamma = (\pi, \tau)$ be a PFS with $\pi \notin \Sigma^*$. Then Γ is reduced if and only if*

$$A_\pi \not\preccurlyeq_p A_\tau^* (A_\tau \neq \varepsilon) \quad \text{or} \quad B_\pi \not\preccurlyeq_s B_\tau^* (B_\tau \neq \varepsilon).$$

Proof. We first show "only if" part. The assertion (i) of Lemma 14, i.e., Γ, is reduced implies the assertion (ii) of Lemma 14. Appealing to Lemma 13, we have the above equation. We apply Lemmas 13 and 12(i) to the above equation then we obtain Γ is reduced. □

Note that by Lemmas 11 and 13(i), $A_\pi \not\preccurlyeq_p A_\tau^* (A_\tau \neq \varepsilon)$ if and only if $A_\pi \not\preccurlyeq_p A_\xi$ for all $\xi \in \Gamma_\tau$.

By the definitions of $w \preccurlyeq_p u^*$, if $A_\pi \not\preccurlyeq_p A_\tau^*$ and $A_\tau \neq \varepsilon$, there is an integer $\hat{i} \geqslant 0$, a pair (a, b) of constant symbols and a triplet (A_0, A_1, A_2) of strings in Σ^* such that

$$a \neq b, \quad A_\tau = A_0 a A_1, \quad A_\pi = A_\tau^{\hat{i}} A_0 b A_2.$$

A tuple $[\hat{i}, (a, b), (A_0, A_1, A_2)]$ is called the p-*tuple* for $A_\pi \not\preccurlyeq A_\tau^* (A_\tau \neq \varepsilon)$. Similarly, we define the s-*tuple* $[\hat{j}, (c, d), (B_0, B_1, B_2)]$ for $B_\pi \not\preccurlyeq_s B_\tau^* (B_\tau \neq \varepsilon)$. By the above theorem, if $\Gamma = (\pi, \tau)$ is reduced, there is the p-tuple for $A_\pi \not\preccurlyeq_p A_\tau^* (A_\tau \neq \varepsilon)$, or the s-tuple for $B_\pi \not\preccurlyeq_p B_\tau^* (B_\tau \neq \varepsilon)$.

By the above theorem, it follows immediately that:

Corollary 17. *For a given PFS $\Gamma = (\pi, \tau)$, the decision problem of whether Γ is reduced is computable in time* $O(|\pi| + |\tau|)$.

4.2. Results on inclusion problem for PFS languages

The following paragraph considers the inclusion problem $L(\Gamma') \subseteq L(\Gamma)$ for given reduced PFSs Γ, Γ'. Before that, we prepare some lemmas.

We first consider the inclusion relation $L(\gamma) \subseteq L(\Gamma)$ for a pattern γ and a PFS Γ.

Lemma 18. *Let $\Gamma = (\pi, \tau)$ be a reduced PFS with $\pi \notin \Sigma^*$ and let γ be a pattern. If $L(\gamma) \subseteq L(\Gamma)$, then either $L(\gamma) \subseteq L(\pi)$ or $L(\gamma) \subseteq L(\Gamma_\tau)$ hold.*

Proof. If $\gamma \in \Sigma^*$, clearly our lemma is valid. Let $\gamma \notin \Sigma^*$. Assume that $L(\gamma) \subseteq L(\Gamma)$, but $L(\gamma) \cap L(\pi) \neq \emptyset$ and $L(\gamma) \cap L(\xi) \neq \emptyset$ for some $\xi \in \Gamma_\tau$. Then by Lemma 12(i), it follows that $(A_\gamma, A_\pi), (A_\gamma, A_\xi) \in \mathrm{Pref}$ and $(B_\gamma, B_\pi), (B_\gamma, B_\xi) \in \mathrm{Suff}$.

Claim. *$A_\pi \preccurlyeq_p A_\gamma$ if $A_\pi \not\preccurlyeq_p A_\tau^*(A_\tau \neq \varepsilon)$, and $B_\pi \preccurlyeq_s B_\gamma$ if $B_\pi \not\preccurlyeq_s B_\tau^*(B_\tau \neq \varepsilon)$.*

The proof of Claim. Assume that $A_\pi \not\preccurlyeq_p A_\tau^*(A_\tau \neq \varepsilon)$ but $A_\pi \not\preccurlyeq_p A_\gamma$. Then by $(A_\gamma, A_\pi), (A_\gamma, A_\xi) \in \mathrm{Pref}$ and Lemma 11, $A_\pi \not\preccurlyeq_p A_\gamma$ implies

$$(*)\ A_\gamma \preccurlyeq_p A_\pi, \quad A_\gamma \preccurlyeq_p A_\xi = (A_\tau)^{i_0} A_\pi$$

for some $i_0 \geqslant 1$. Let $[\hat{i}, (a, b), (A_0, A_1, A_2)]$ be the p-tuple for $A_\pi \not\preccurlyeq_p A_\tau^*(A_\tau \neq \varepsilon)$. By the definition of the p-tuple, we obtain $A_\tau = A_0 a A_1$, $a \neq b$ and

$$A_\pi = A_\tau^{\hat{i}} A_0 b A_2, \quad A_\xi = (A_\tau)^{\hat{i}+i_0} A_0 b A_2 = A_\tau^{\hat{i}} (A_0 a A_1)(A_\tau)^{i_0-1} A_0 b A_2.$$

Then by $(*)$ and $a \neq b$, it follows that $A_\gamma \preccurlyeq_p A_\tau^{\hat{i}} A_0$. Let a_1 and a_2 be the $(|A_\gamma|+1)$st constant symbols from the left side of A_π and A_ξ, respectively. Since $\sharp\Sigma \geqslant 3$, there is a symbol $c \in \Sigma$ such that $c \neq a_j$ for $j = 1, 2$. Let x be the left-most variable of γ and let $\gamma_c = \gamma\{x := cx\}$. Then by the choice of c, clearly $A_{\gamma_c} \not\preccurlyeq_p A_\tau^j A_\pi$ holds for all $j \geqslant 0$. By Lemmas 11 and 12(i), it means that $L(\gamma_c) \cap L(\pi) = \emptyset$ and $L(\gamma_c) \cap L(\Gamma) = \emptyset$. Since $L(\gamma_c) \subseteq L(\gamma)$, it contradicts that $L(\gamma) \subseteq L(\Gamma)$. Therefore $A_\pi \preccurlyeq_p A_\gamma$ is valid.

Similarly, we can prove $B_\pi \preccurlyeq_s B_\gamma$ if $B_\pi \not\preccurlyeq_s B_\tau^*(B_\tau \neq \varepsilon)$. This completes the proof of Claim.

Since Γ is reduced, by Theorem 16, $A_\pi \not\preccurlyeq_p A_\tau^*(A_\tau \neq \varepsilon)$ or $B_\pi \not\preccurlyeq_s B_\tau^*(B_\tau \neq \varepsilon)$ holds.

If $A_\pi \not\preccurlyeq_p A_\tau^*(A_\tau \neq \varepsilon)$, then $A_\pi \not\preccurlyeq_p (A_\tau)^{i_0} A_\pi (= A_\xi)$, but by Claim, $A_\pi \preccurlyeq_p A_\gamma$. Since $|A_\pi| < |A_\gamma|$, it contradicts that $(A_\gamma, A_\xi) \in \mathrm{Pref}$.

Similarly we can prove for the case of $B_\pi \not\preccurlyeq_s B_\tau^*(B_\tau \neq \varepsilon)$. This completes our proof. □

Next, we consider the inclusion relation $L(\gamma) \subseteq L(\Gamma_\tau)$ for a pattern γ and a PFS Γ. We give a relationship between τ_π and a pattern $\xi \in \Gamma_\tau$ for a PFS Γ, where τ_π is defined in Section 2. Note that the pattern τ_π is the shortest pattern in the set Γ_τ and $A_{\tau_\pi} = A_\tau A_\pi$ and $B_{\tau_\pi} = B_\pi B_\tau$ hold.

Lemma 19. *Let $\Gamma = (\pi, \tau)$ be a PFS with $\pi \notin \Sigma^*$ and let $\xi \in \Gamma_\tau$. Then*

$$L(\xi) \cap L(\tau_\pi) \neq \emptyset \iff \xi \preccurlyeq \tau_\pi.$$

Proof. We prove the "only if" part since the "if" part is obvious. Assume that $L(\xi) \cap L(\tau_\pi) \neq \emptyset$ for $\xi \in \Gamma_\tau$ with $\xi \neq \tau_\pi$. Since $|A_{\tau_\pi}| \leqslant |A_\xi|$ and $|B_{\tau_\pi}| \leqslant |B_\xi|$ hold, by Lemma 12(i), it leads that $A_{\tau_\pi} \preccurlyeq_p A_\xi$ and $B_{\tau_\pi} \preccurlyeq_s B_\xi$. By the definition of Γ_τ, ξ' contains the pattern γ as a substring, where $\xi = (A_\xi x)\xi'(x' B_\xi)$ and $\tau_\pi = (A_{\tau_\pi} x)\gamma(x' B_{\tau_\pi})$, and thus by Lemma 12(ii), $L(\xi) \subseteq L(\tau_\pi)$, that is, $\xi \preccurlyeq \tau_\pi$. □

Using the above lemma, it follows that

Lemma 20. *Let $\Gamma = (\pi, \tau)$ be a reduced PFS and let γ be a pattern with $L(\gamma) \subseteq L(\Gamma_\tau)$. Then*
(i) *$A_{\tau_\pi} \preccurlyeq_p A_\gamma$ if $A_\pi \not\preccurlyeq_p A_\tau^*(A_\tau \neq \varepsilon)$, and $B_{\tau_\pi} \preccurlyeq_s B_\gamma$ if $B_\pi \not\preccurlyeq_s B_\tau^*(B_\tau \neq \varepsilon)$.*
(ii) *$L(\gamma) \cap L(\tau_\pi) \neq \emptyset \iff L(\gamma) \subseteq L(\tau_\pi)$.*

The proof can be done similarly to that of Lemma 18. The assertion (i) corresponds to Claim in the proof of Lemma 18.

By Lemmas 18 and 20, we have the following results of the inclusion relation for reduced PFSs:

Theorem 21. *Let $\Gamma = (\pi, \tau)$ and $\Gamma' = (\alpha, \beta)$ be reduced PFSs with $\pi, \alpha \notin \Sigma^*$. If $L(\Gamma') \subseteq L(\Gamma)$, the following equivalences are valid.*
(i) *$L(\Gamma') \subseteq L(\pi) \iff \alpha \preccurlyeq \pi, \beta_\alpha \preccurlyeq \pi$,*
(ii) *$L(\alpha) \subseteq L(\pi), L(\Gamma'_\beta) \subseteq L(\Gamma_\tau) \iff \alpha \preccurlyeq \pi, \beta_\alpha \not\preccurlyeq \pi$.*

Proof. Since the "only if" parts of two assertions are obvious, we only prove the "if" parts.

Since Γ' is reduced, by Theorem 14, $L(\alpha) \cap L(\Gamma'_\beta) = \emptyset$ holds.

(i) We assume that $L(\Gamma') \subseteq L(\Gamma)$, $\alpha \preccurlyeq \pi$ and $\beta_\alpha \preccurlyeq \pi$ but $L(\Gamma') \not\subseteq L(\pi)$. Then since $\alpha \preccurlyeq \pi$ implies $L(\alpha) \subseteq L(\pi)$, it means that $L(\Gamma'_\beta) \not\subseteq L(\pi)$, and thus $L(\gamma) \not\subseteq L(\pi)$ for some $\gamma \in \Gamma'_\beta$. Since $L(\gamma) \subseteq L(\Gamma)$, by Lemma 18, we have $L(\gamma) \subseteq L(\Gamma_\tau)$. Since $\beta_\alpha \preccurlyeq \pi$, by Lemma 12(i), $A_\pi \preccurlyeq_p A_{\beta_\alpha} (= A_\beta A_\alpha)$ and $B_\pi \preccurlyeq_s B_{\beta_\alpha} (= B_\alpha B_\beta)$ hold. Since $L(\gamma) \cap L(\pi) = \emptyset$ and $\gamma \in \Gamma'_\beta$, by Lemma 11, we have $A_\gamma = A_\beta^{i_0} A_\alpha$ and $B_\alpha B_\beta^{j_0} (= B_\gamma)$. By Theorem 16 and Lemma 18, it follows that

$$(*)\quad A_\pi \not\preccurlyeq_p A_\beta^{i_0} A_\alpha (= A_\gamma) \quad \text{or} \quad B_\pi \not\preccurlyeq_s B_\alpha B_\beta^{j_0} (= B_\gamma)$$

for some $i_0, j_0 \geqslant 1$. Since $\alpha \preccurlyeq \pi$ and $\beta_\alpha \preccurlyeq \pi$, if $A_\alpha \not\preccurlyeq_p A_\beta^* (A_\beta \neq \varepsilon)$ then $A_\pi \preccurlyeq_p A_\beta^*$, Otherwise we have $A_\pi \preccurlyeq_p A_\beta A_\alpha$, and $A_\alpha \preccurlyeq_p A_\beta^*$ or $A_\beta = \varepsilon$. In any way, we have $A_\pi \preccurlyeq_p A_\gamma$. Similarly we have $B_\pi \preccurlyeq_s B_\gamma$. It contradicts $(*)$. Therefore $L(\Gamma'_\beta) \subseteq L(\pi)$ holds.

(ii) We assume that $L(\Gamma') \subseteq L(\Gamma)$, $\alpha \preccurlyeq \pi$ and $\beta_\alpha \not\preccurlyeq \pi$. Then by $\alpha \preccurlyeq \pi$, $L(\alpha) \subseteq L(\pi)$ holds. Since $\beta_\alpha \not\preccurlyeq \pi$, by Lemma 18, we have $L(\beta_\alpha) \cap L(\pi) = \emptyset$. It means that $A_\pi \not\preccurlyeq_p A_{\beta_\alpha}$ or $B_\pi \not\preccurlyeq_s B_{\beta_\alpha}$.

Suppose $L(\Gamma'_\beta) \not\subseteq L(\Gamma_\tau)$ and $L(\gamma) \not\subseteq L(\Gamma_\tau)$ for some $\gamma \in \Gamma_\beta$. Then since $L(\Gamma_\beta) \subseteq L(\Gamma)$, it follows that $L(\gamma) \cap L(\pi) \neq \emptyset$. By Lemma 18, it means that $\gamma \preccurlyeq \pi$. Hence, $A_\pi \preccurlyeq_p A_\gamma$ and $B_\pi \preccurlyeq_s B_\gamma$ hold. Similarly to the proof of (i), it implies that $A_\pi \preccurlyeq_p A_{\beta_\alpha}$ and $B_\pi \preccurlyeq_s B_{\beta_\alpha}$, and a contradiction. □

5. Learnability of erasing PFSs

In the present section, we show that the class $\mathcal{PFSL}$ is inferable from positive examples. In Section 3, we have proved that the class has M-finite thickness. Thus by Theorem 9, it is enough to show that each PFS language has a finite tell tale set.

5.1. Finite tell tale sets for PFS languages

Let $\Gamma = (\pi, \tau)$ be a PFS. We introduce the following particular finite subset of $L(\Gamma)$:

$$T(\Gamma) = L(\Gamma) \cap \Sigma^{\leqslant |\tau_{\tau_\pi}|},$$

where $\Sigma^{\leqslant l}$ denotes the set of strings with length at most $l \geqslant 0$. We prove that the above set is a finite tell tale set of $L(\Gamma)$. The methods of the proof differ according to the cases of $\pi \notin \Sigma^*$, $\pi \in \Sigma^+$ and $\pi = \varepsilon$. For the first case, the inclusion theorem given in the previous section plays an important role in the proof. For the other cases, it is necessary to investigate some combinatorial properties on constant strings.

Case of $\pi \notin \Sigma^*$: For a pattern π, $S(\pi)$ denotes the set of all strings obtained from π by substituting the empty string or a constant symbol in Σ to each variable.

The lemma below is fundamental to discuss a finite tell tale for a PFS language in this case.

Lemma 22 (*Uemura and Sato [19]*). *Suppose that* $\sharp\Sigma \geqslant 3$. *Let* π *and* τ *be erasing regular patterns. Then*

$$S(\pi) \subseteq L(\tau) \iff \pi \preccurlyeq \tau.$$

If a PFS $\Gamma = (\pi, \tau)$ is not reduced, clearly $L(\Gamma) = L(\pi) = L(\pi, x)$ holds.

Theorem 23. *Let* $\Gamma = (\pi, x)$ *be a PFS with* $\pi \notin \Sigma^*$. *Then the set* $T(\Gamma)$ *is a finite tell tale set of the language* $L(\Gamma)$.

Proof. We prove by contraposition. Assume that $T(\Gamma) \subseteq L(\Gamma') \subsetneq L(\pi) (= L(\pi, x))$ for some PFS $\Gamma' = (\alpha, \beta)$. Since $S(\pi) \subseteq T(\Gamma)$, by Lemma 22, Γ' must be reduced, that is, $L(\alpha) \subsetneq L(\alpha) \cup L(\Gamma'_\beta) (= L(\Gamma'))$. By our assumption, it follows that $\alpha \preccurlyeq \pi$ but $\alpha \neq \pi$, and $\beta_\alpha \preccurlyeq \pi$. Moreover, since $c(\pi)$ is the unique shortest string of $S(\pi)$ and $L(\pi)$, $c(\pi) = c(\alpha)$ holds. It means that $\alpha \preccurlyeq \alpha' x \alpha'' \preccurlyeq \pi$ for some α', α'' such that $\alpha = \alpha'\alpha''$.

Case of $\alpha', \alpha'' \neq \varepsilon$: Let a and b be the $|c(\alpha')|$th and $(|c(\alpha')| + 1)$st constant symbols of $c(\alpha)$, respectively. Since $\sharp\Sigma \geqslant 3$, there is a symbol $c \in \Sigma$ different from a and b. Consider the string $w = w'cw''$ with length $|c(\alpha)| + 1$, where

$w' = c(\alpha')$ and $w'' = c(\alpha'')$. Clearly $w \in S(\pi)$ but $w \notin L(\alpha)$, and thus $w \in L(\beta_\alpha)$. It implies that $\beta = dx$ or xd for some $d \in \Sigma$. If $\beta = dx$, then $w = w'cw'' = dw'w''$, and so $w'c = dw'$. It leads that $c = d$ and $w' \in \{c\}^+$, and thus $a = c$. It contradicts the choice of c.

Similarly we prove for the cases of $\alpha' = \varepsilon$ or $\alpha'' = \varepsilon$. □

Theorem 24. *Let $\Gamma = (\pi, \tau)$ be a reduced PFS with $\pi \notin \Sigma^*$. Then the set $T(\Gamma)$ is a finite tell tale set of the language $L(\Gamma)$.*

Proof. We prove by contraposition. Assume that $T(\Gamma) \subseteq L(\Gamma') \subsetneq L(\Gamma)$ for some PFS $\Gamma' = (\alpha, \beta)$.

(i) *Case that Γ' is not reduced, that is, $L(\Gamma') = L(\alpha)$*: In this case, it follows that

$$S(\pi) \subseteq T(\Gamma) \subseteq L(\alpha) \subsetneq L(\pi) \cup L(\Gamma_\tau) = L(\Gamma).$$

By Lemma 22, it follows that $\pi \preccurlyeq \alpha$. Since $L(\alpha) \cap L(\pi) \neq \emptyset$, by Lemma 18, it follow that $L(\alpha) \subseteq L(\pi)$, and thus $L(\alpha) = L(\pi)$. Since Γ is reduced, $c(\tau_\pi)$ belongs to $L(\Gamma_\tau)$ but not $L(\pi)$. By the definition of $T(\Gamma)$, $c(\tau_\pi) \in T(\Gamma)$. It contradicts $T(\Gamma) \subseteq L(\alpha)(= L(\pi))$.

(ii) *Case that Γ' is reduced*: Clearly $c(\pi)$ and $c(\alpha)$ are the shortest strings of $L(\Gamma)$ and $L(\Gamma')$, respectively, and by the above assumption, $c(\pi) = c(\alpha)$, and so $L(\alpha) \cap L(\pi) \neq \emptyset$. By Lemma 18, it implies that $\alpha \preccurlyeq \pi$. Since $L(\Gamma') \subsetneq L(\Gamma)$, by Theorem 21(i), $\beta_\alpha \preccurlyeq \pi$ implies $L(\Gamma') \subseteq L(\pi)$, and so $c(\tau_\pi) \in L(\pi)$. It is a contradiction since Γ is reduced and thus $L(\pi) \cap L(\Gamma_\tau) = \emptyset$ holds. Hence $\beta_\alpha \not\preccurlyeq \pi$ holds. By Theorem 21(ii), it follows that $L(\alpha) \subseteq L(\pi)$ and $L(\Gamma'_\beta) \subseteq L(\Gamma_\tau)$, and thus

$$(*)\quad S(\pi) \subseteq L(\alpha), \quad S(\tau_\pi) \subseteq L(\Gamma'_\beta) \subseteq L(\Gamma_\tau).$$

By the former of $(*)$ and Lemma 22, we have $\pi \preccurlyeq \alpha$, and thus $\pi \equiv \alpha$. Since both π and α are of canonical form, it implies that $\pi = \alpha$.

Let us prove $\beta = \tau$. We should note that the shortest strings of $L(\Gamma_\tau)$ and $L(\Gamma'_\beta)$ are $c(\beta_\pi)$ and $c(\tau_\pi)$, respectively. By the latter of $(*)$, it implies that $c(\beta_\pi) = c(\tau_\pi)$. Since $L(\beta_\alpha) \subseteq L(\Gamma_\tau)$ and $L(\beta_\alpha) \cap L(\tau_\pi) \neq \emptyset$, by Lemma 20, it follows that $L(\beta_\alpha) \subseteq L(\tau_\pi)$, and so $\beta_\pi \preccurlyeq \tau_\pi$. Similar to the proof of Lemma 20, $\tau_\pi \not\preccurlyeq \beta_\pi$ is shown to imply $L(\tau_\pi) \cap L(\beta_\pi) = \emptyset$, and a contradiction. Therefore $\tau_\pi \preccurlyeq \beta_\pi$ holds. Consequently, we obtain $\beta_\pi \equiv \tau_\pi$. In terms of Corollary 15, both β_π and τ_π are of *canonical* form, and thus $\beta_\pi = \tau_\pi$. Since π contains variables, it is easily shown that $\beta = \tau$.

It means that $L(\Gamma') = L(\Gamma)$ holds, and a contradiction. □

Case of $\pi \in \Sigma^+$: Here we should note that $\pi \in \Sigma^*$ implies $\Gamma_\tau \subseteq \Sigma^*$ and so $L(\Gamma_\tau) = \Gamma_\tau$. Let us put $w = \pi$. First we will prepare some lemmas about multiple strings.

Let $u' = \rho(w, u)$ for $w \preccurlyeq_p u^*$ and $u = u'u''$ for some u''. Then since $(u'u'')^i u' = w = u'(u''u')^i$, it implies that $w \preccurlyeq_p u^*$ if and only if $w \preccurlyeq_s v^*$ for $v = u''u'$. Hence, we have $w \in \{u\}^+ \iff \rho_p(w, u) = \varepsilon \iff w \preccurlyeq_p u^*$ and $w \preccurlyeq_s u^*$. Moreover, $w_1 w_2 \in \{u\}^+$ implies $w_1 \rho_s(w_2, u), \rho_p(w_1, u) w_2 \in \{u\}^+$.

The following result plays an essential role in our problem on finite tell tale sets.

Lemma 25. *Let w, u_1, u_2, u_3 and v be nonempty strings in Σ^+.*

(i) *If $w = u_1u_2 = u_2u_1$, then w is a multiple string and $w, u_i \in \{u\}^+$ holds for some nonmultiple string $u \in \Sigma^+$.*

(ii) *If $u_1 v = v u_2$, then $v \preccurlyeq_p u_1^*$, $v \preccurlyeq_s u_2^*$ and $u_2 = u''u'$, where $u' = \rho_p(v, u_1)$ and $u_1 = u'u''$ for some $u'' \in \Sigma^+$.*

(iii) *If $u_1(wu_2) = (wu_2)u_1$, then $w \preccurlyeq_p u^*$, $u_1(wu_2)w \preccurlyeq_p u^*$ and $\rho_p(w, u) = \rho_p(u_1(wu_2)w, u)$ for some nonmultiple string $u \in \Sigma^+$.*

(iv) *If $(u_1w)u_2 = u_2(u_1w)$, then $w \preccurlyeq_s u^*$, $w(u_1w)u_2 \preccurlyeq_s u^*$ and $\rho_s(w, u) = \rho_s(w(u_1w)u_2, u)$ for some nonmultiple string $u \in \Sigma^+$.*

(v) *If $wu \in \{u_1\}^+$ and $uw \in \{u_2\}^+$ for nonmultiple strings u_1 and u_2, then $u_2 = u''u'$ holds, where $u' = \rho_p(w, u_1)$ and $u_1 = u'u''$ for some $u'' \in \Sigma^+$.*

(vi) *If $u_1(wu_2) = (wu_2)u_1$ and $(u_2w)u_3 = u_3(u_2w)$, then $w \preccurlyeq_p u^*$, $u_1wu_2wu_3 \preccurlyeq_p u^*$ and $\rho_p(w, u) = \rho_p(u_1wu_2 wu_3, u)$ for some nonmultiple string $u \in \Sigma^+$.*

(vii) *If $w \in \{u\}^+$ for nonmultiple u and $u_1 w = w u_2$ with $|u_1| \leqslant |w|$, then $u_i \in \{u\}^*$ for $i = 1, 2$.*

Proof. (i) Let $w = u_1u_2 = u_2u_1$ for some $u_1, u_2 \in \Sigma^+$. Then since $w \preccurlyeq_p u_1 w (= wu_1)$ and $w \preccurlyeq_s u_2 w (= wu_2)$, by Lemma 13, it implies that $w \preccurlyeq_p u_1^*$ and $w \preccurlyeq_s u_2^*$. Let u be the nonmultiple string satisfying $u_1 \in \{u\}^+$. Then clearly $w \preccurlyeq_p u^*$ holds. Let $u' = \rho_p(w, u)$ and $u = u'u''$ for some $u'' \in \Sigma^+$.

Let $|u_1| \geqslant |u_2|$ and we take two integers k_1 and l_1 such that $|u_1| = k_1 \cdot |u_2| + l_1$, $k_1 \geqslant 1$ and $0 \leqslant l_1 < |u_2|$. Then by $u_1u_2 = u_2u_1$, it can be easily seen that $u_1 = u_2^{k_1}u_3$ and $u_2u_3 = u_3u_2$, where u_3 is the suffix of u_1 with length l_1. If $l_1 = |u_3| = 0$, then the string w is multiple, and moreover $w, u_i \in \{u_2\}^+$. Otherwise $|u_2| > |u_3| > 0$. By repetition of the above, we obtain the following sequence $u_1, u_2, \ldots$ such that

$$u_i = u_{i+1}^{k_i}u_{i+2}, \quad k_i \geqslant 1, \quad u_{i+2} \neq \varepsilon, \quad u_iu_{i+1} = u_{i+1}u_i$$

for $i \geqslant 1$. Since $|u_1| > |u_2| > \cdots$, there is an integer $t \geqslant 3$ satisfying $u_t \neq \varepsilon$ but $u_{t+1} = \varepsilon$. As easily seen, w is a multiple string of the nonempty string u_t, and $w, u_i \in \{u_t\}^+$.

The assertion (ii) can be similar to the above.

(iii) By the assertion (i), $u_1, wu_2 \in \{u\}^+$ hold for some nonmultiple string $u \in \Sigma^+$. It means that $w \preccurlyeq_p u^*$, and so $u_1(wu_2)w \preccurlyeq_p u^*$ and $\rho_p(w, u) = \rho_p(u_1(wu_2)w, u)$.

The assertion (iv) can be shown similarly to the assertion (iii).

(v) Let $wu \in \{u_1\}^+$ and $uw \in \{u_2\}^+$ for multiple strings u_1 and u_2. Without loss of generality, we assume that $|u_1| \leqslant |u_2|$.

Let $u' = \rho_p(w, u_1)$ and $u_1 = u'u''$ for some $u'' \in \Sigma^+$. Then $w = u_1^iu'$ and $u = u''u_1^j$ for some $i, j \geqslant 0$. Since $uw = u''u_1^{i+j}u' = (u''u')^{i+j+1}$ and $uw \in \{u_2\}^+$, $v^{i+j+1} = u_2^l$ holds for some $l \geqslant 1$, where $v = u''u'$.

Assume that $(|v| =)|u_1| < |u_2|$, and thus $v \neq u_2$. Since u_2 is nonmultiple, $l \geqslant 2$. Indeed, if $l = 1$ then $v = u_2$. Let $v' = \rho_p(u_2, v)$ and $v = v'v''$ for some $v'' \in \Sigma^+$. Then $u_2 = v^kv'$ holds for some $k \geqslant 1$. If $v' = \varepsilon$ then $u_2 = v^k$. Since $|v| < |u_2|$, it means that $k \geqslant 2$, and a contradiction because of u_2 to be nonmultiple. Hence $v' \neq \varepsilon$ holds. Since $l \geqslant 2$, $u_2^2 \preccurlyeq_p v^*$ holds. It means that $v'v = v'(v'v'') \preccurlyeq_p v^*$, and thus $v'(v'v'') = vv' = (v'v'')v'$. Therefore we have $v'v'' = v''v'$. Since $v', v'' \in \Sigma^+$, by the assertion (i), it implies that $v', v'' \in \{\hat{u}\}^+$ for some nonmultiple string $\hat{u}$. It means that u_2 is nonmultiple, and a contradiction.

Hence we have $|u_1| = |u_2|$, and thus $u_2 = u''u'$.

(vi) By the assertion (i), $u_1, wu_2 \in \{u_1'\}^+$ and $u_2w, u_3 \in \{u_2'\}^+$ hold for some nonmultiple strings $u_1', u_2' \in \Sigma^+$. It implies that $w \preccurlyeq_p {u_1'}^*$ and $w \preccurlyeq_s {u_2'}^*$, and by the assertion (v), we have $u_1' = u'u''$ and $u_2' = u''u'$, where $u' = \rho_p(w, u_1')$ and $u_1' = u'u''$ some $u'' \in \Sigma^+$. Hence we have $u_1wu_2wu_3 \preccurlyeq_p {u_1'}^*$.

The proof of the assertion (vii) can be easily derived from the assertion (vi). □

Now we consider some specific strings contained in $T(\Gamma)$ for $\Gamma = (w, \tau)$ with $w \in \Sigma^+$. By the definition of $\mathcal{PFS}$ given in Section 2, the strings w and τ_w satisfy either one of $\tau_w \not\preccurlyeq_p u^*$ and $\rho_p(\tau_w, u) \neq w$ for any nonmultiple string u with $|u| < |\tau_w|$. Hence, it follows that for any nonmultiple string u with $|u| < |\tau_w|$,

(i) $\tau_w \not\preccurlyeq_p u^*$ or $w \not\preccurlyeq_p u^*$ or $\rho_p(\tau_w, u) \neq \rho_p(w, u)$ or

(ii) $w, \tau_w \preccurlyeq_p u^*$ and $\rho_p(\tau_w, u) = \rho_p(w, u)$ but $|u| \leqslant |w|$.

Let $\tau = v_0x_1v_1x_2 \ldots x_nv_n$ for some $v_i \in \Sigma^*$ $(i = 0, \ldots, n)$. Then we have $\tau_w = v_0wv_1w \ldots wv_n$. We introduce the following strings:

$$r_i(\tau) = \tau\{x_i := \tau_w, x_j := w \mid j \neq i\}, \quad i = 1, \ldots, n.$$

That is, $r_i(\tau) = v_0wv_1 \ldots wv_{i-1}((v_0wv_1 \ldots wv_{i-1})w(v_iw \ldots wv_n))v_iw \ldots wv_n$. By the definition of $T(\Gamma)$, we have

$$\{w, \tau_w\} \cup \{r_i(\tau) \mid i = 1, \ldots, n\} \subseteq T(\Gamma).$$

Note that the strings $w, \tau_w, r_i(\tau)$s are the unique shortest, the unique second shortest and the third shortest strings of $L(w, \tau)$ and $T(\Gamma)$.

We first introduce the following condition for the particular strings w and $\tau_w \in T(\Gamma)$ that corresponds to the above (ii):

Condition A $\tau_w \preccurlyeq_p u^*$, $w \preccurlyeq_p u^*$ and $\rho_p(\tau_w, u) = \rho_p(w, u)$ for some nonmultiple string u with $|u| \leqslant |w|$.

Then we show that if Condition A is satisfied, then $L(w,\tau) = L(w, u^k x)$ for some $k \geqslant 1$. That is, the language $L(w,\tau)$ depends only on the string w, the nonmultiple string u and the integer k determined by $|w|$ and $|\tau_w|$ as shown below. Otherwise, we show that the $r_i(\tau)$s are mutually distinct if $v_0 v_n \neq \varepsilon$, and so except $r_1(\tau) = r_n(\tau)$ otherwise.

Lemma 26. *Let $\Gamma = (w,\tau)$ be a reduced PFS with $w \in \Sigma^+$ and $var(\tau) = \{x_1,\ldots,x_n\}$ $(n \geqslant 1)$. If Condition A is satisfied, then $r_i(\tau) = r_j(\tau)$ for any i, j, and $L(\Gamma) = L(w, u^k x)$ where k is a positive integer such that $u^k w = \tau_w$.*

Proof. Let $u' = \rho_p(\tau_w, u) = \rho_p(w,u)$ and $u = u'u''$ for some $u'' \in \Sigma^+$. Then $w = u^{l_0}u'$ and $\tau_w = u^l u'$ hold for some $l_0, l \geqslant 1$. By Condition A, the following holds:

$$v_0 \in \{u\}^*, \quad wv_i \in \{u\}^+ \ (i = 1,\ldots,n-1), \quad wv_n = u^m u' \text{ for some } m \geqslant 1.$$

In fact, since $\tau_w = v_0 w V \preccurlyeq_p u^*$ for $V = v_1 w \ldots w v_n$, we have $v_0 \preccurlyeq_p u^*$ and $v_0 w \preccurlyeq_p u^*$. If $v_0 \notin \{u\}^*$, then $v' = \rho_p(v_0,u) \neq \varepsilon$ and $v'w (= v'u^j u') \preccurlyeq_p u^*$ for some j. Since $l_0 \geqslant 1$, it implies that $v'u \preccurlyeq_p u^*$ and thus $v'v'' = v''v'$ for some nonempty string v'' with $u = v'v''$. Appealing to Lemma 25(i), v', $v'' \in \{\hat{u}\}^+$. It means that u is multiple, and a contradiction. Similarly it can be shown that $wv_i \in \{u\}^+$ for each i.

By the above, we have $r_i(\tau) = u^{2l-l_0}u'$ for every i, and thus $r_i(\tau) = r_j(\tau)$ for any i, j.

Let us put $\Gamma'' = (w, u^k x)$ where $k = l - l_0$. As easily seen, $L(\Gamma'') = \{u^{ik+l_0}u' \mid i \geqslant 0\}$. We show that $L(\Gamma) = L(\Gamma'')$. Clearly, w is the shortest string of $L(\Gamma'')$. Remember that $L(\Gamma_\tau) = \Gamma_\tau = \bigcup_{t=1}^{\infty} \Gamma_t$. We first show by mathematical induction on $t \geqslant 1$ that $\Gamma_t \subseteq L(\Gamma'')$. Since $\Gamma_1 = \{\tau_w\}$ and $\tau_w = u^l u'$, it is valid. Assume that $\Gamma_t \subseteq L(\Gamma'')$ for $t \leqslant t_0 - 1$. Let $\xi \in L(\Gamma_t)$. Then we have $\xi = \tau\{x_1 := \xi_1,\ldots,x_n := \xi_n\}$ for some $\xi_j \in \Gamma_{t_0-1}$ $(j = 1,\ldots,n)$. By induction hypothesis, $\xi_j = u^{m_j \cdot k + l_0}u'$ holds for some $m_j \geqslant 0$ $(j = 1,\ldots,n)$, and thus $\xi = v_0(u^{m_1\cdot k+l_0}u'v_1)(u^{m_2\cdot k+l_0}u'v_2)\cdots(u^{m_n\cdot k+l_0}u'v_n)$ holds. For $j = 1,\ldots,n-1$, since $u^{m_j\cdot k+l_0}u'v_j = u^{m_j\cdot k}(wv_j)$ and $wv_j \in \{u\}^+$ by $(*)$, it implies that $u^{m_j\cdot k+l_0}u'v_j \in \{u\}^+$. Similarly $u^{m_n\cdot k+l_0}u'v_n = u^{m_n\cdot k+l'}u'$ holds. Since $v_0 \in \{u\}^*$, we finally obtain $\xi = u^{m\cdot k+l_0}u'$, where $m = l + m_1 + m_2 + \cdots + m_n$. Hence we have $\xi \in L(\Gamma'')$. Therefore, $L(\Gamma) \subseteq L(\Gamma')$ holds.

Conversely, we can prove by mathematical induction on $i \geqslant 0$ that $u^{ki+l_0}u' \in L(\Gamma)$. Therefore we obtain $L(\Gamma) = L(\Gamma'')$. Since $L(\Gamma') = L(\Gamma'')$, it follows that $L(\Gamma) = L(\Gamma')$. □

By Lemma 25, it follows that:

Lemma 27. *Let $\Gamma = (w,\tau)$ be a reduced PFS with $w \in \Sigma^+$ and $var(\tau) = \{x_1,\ldots,x_n\}$ $(n \geqslant 2)$. If Condition A is not satisfied, then the $r_i(\tau)$s are mutually distinct if $v_0 v_n \neq \varepsilon$, and so except $r_1(\tau) = r_n(\tau)$ otherwise.*

Proof. Assume that $r_i(\tau) = r_j(\tau)$ for some i, j such that $1 \leqslant i < j \leqslant n$. Then $\tau_w = V_1 w V_2 w V_3$, and

$$(*)\ V_1(wV_2) = (wV_2)V_1, \quad V_3(V_2 w) = (V_2 w)V_3,$$

where $V_1 = v_0 w v_1 \ldots w v_{i-1}$, $V_2 = v_i w \ldots w v_{j-1}$ and $V_3 = v_j w \ldots w v_n$.

Note that $V_1 = \varepsilon \iff i = 1$ and $v_0 = \varepsilon$, and $V_3 = \varepsilon \iff j = n$ and $v_n = \varepsilon$. Moreover, by Lemma 25(i) and $(*)$, V_1, $wV_2 \in \{u_1\}^+$ if $V_1 \neq \varepsilon$, and $V_2 w$, $V_3 \in \{u_2\}^+$ if $V_3 \neq \varepsilon$ for some nonmultiple strings $u_1, u_2 \in \Sigma^+$.

Case of $V_1, V_3 \neq \varepsilon$: In this case, by Lemma 25(vi) and the above, we have $w \preccurlyeq_p u^*$, $\tau_w = V_1 w V_2 w V_3 \preccurlyeq_p u^*$ and $\rho_p(w,u) = \rho_p(\tau_w, u)$ for some nonmultiple string u with $|u| < |\tau_w|$, and a contradiction.

Case of $V_1 \neq \varepsilon$ and $V_3 = \varepsilon$: By Lemma 25(iii) and the above, we have $w \preccurlyeq_p u^*$, $\tau_w \preccurlyeq_p u^*$ and $\rho_p(w,u) = \rho_p(\tau_w,u)$ for some nonmultiple string u with $|u| < |\tau_w|$, and a contradiction. Similarly we can prove for the case of $V_1 = \varepsilon$ and $V_3 \neq \varepsilon$.

Case of $V_1 = V_3 = \varepsilon$: In this case, as mentioned above, it is possible only for $v_0 = v_n = \varepsilon$ and $i = 1$, $j = n$. Clearly $r_1(\tau) = r_n(\tau)$ holds for this case. □

Finally, we prove that $T(\Gamma)$ is a finite tell tale set for $L(\Gamma)$ in this case.

Theorem 28. *Let $\Gamma = (w,\tau)$ be a reduced PFS with $w \in \Sigma^+$. Then the set $T(\Gamma)$ is a finite tell tale set of the language $L(\Gamma)$.*

Proof. We assume that $T(\Gamma) \subseteq L(\Gamma') \subsetneq L(\Gamma)$ for some PFS $\Gamma' = (\alpha,\beta)$.

Claim A. $\alpha = w$ *and* $\beta_\alpha = \tau_w$.

The proof of Claim A. By the assumption of the inclusion, w is the unique shortest string of $L(\Gamma')$, and so $w = c(\alpha)$. We show that α does not contain variables. Suppose that α contains some variables. Without loss of generality, we can assume that Γ' is reduced, and by Corollary 15, $A_\alpha \neq \varepsilon$ or $B_\alpha \neq \varepsilon$. Let $A_\alpha \neq \varepsilon$, and $\alpha = A_\alpha x_1 \alpha'$ for some α'. Since $\sharp\Sigma \geqslant 3$, there is a constant symbol $c \in \Sigma$ different from the $|A_\alpha|$th and $(|A_\alpha|+1)$st symbols of w. Consider a pattern $\alpha_c = \alpha\{x_1 := c\} = A_\alpha c \alpha'$. Since the string $c(\alpha_c)$ has the length $|w|+1$, by $c(\alpha_c) \in L(\Gamma)$, it must equal τ_w which is the second shortest string in $L(\Gamma)$. It means that $\tau = ax$ or xa for some $a \in \Sigma$, and $\tau_w = aw$ or wa. By the choice of c, both of the $(|A_\alpha|+1)$st symbols of wa and aw, however, are different from c. Thus $c(\alpha_c) \neq wa$ and $c(\alpha_c) \neq aw$ hold, and so $c(\alpha_c) \neq \tau_w$ in any case. Therefore, α does not contain any variable and so $\alpha = w$. Since τ_w is the unique second shortest string of $L(\Gamma')$, it implies that $\beta_w = \tau_w$.

Let $\tau = v_0 x_1 v_1 x_2 \dots x_n v_n$ and $\beta = v'_0 y_1 \dots y_{n'} v'_{n'}$ for some $v_i, v'_j \in \Sigma^*(i = 0, \dots, n,\ \ j = 0, \dots, n')$. Clearly $r_i(\tau)$s are all of the third shortest strings of $L(\Gamma)$ and $S(\Gamma)$, and $r_j(\beta)$s are those of $L(\Gamma')$. It means that

$$(*)\ \{r_i(\tau) \mid i = 1, \dots, n\} = \{r_j(\beta) \mid j = 1, \dots, n'\}.$$

If Condition A holds, by Lemma 26, it follows that $L(\Gamma) = L(w, u^k x)$, where k is a positive integer such that $u^k w = \tau_w$. Thus by Claim A, $L(\Gamma') = L(\Gamma)$ holds.

Assume that Condition A does not hold, that is, for any nonmultiple string $u \in \Sigma^+$ with $|u| < |w|$, $\tau_w \not\preccurlyeq_p u^*$, $w \not\preccurlyeq_p u^*$, or $\rho_p(\tau_w, u) \neq \rho_p(w, u)$.Then by Lemma 27, it follows that all $r_i(\tau)$s are distinct if $v_0 v_n \neq \varepsilon$, and so except $r_1(\tau) = r_n(\tau)$, otherwise. Let us put $r_i(\tau) = V_{i,1}\tau_w V_{i,2}$ and $r_j(\beta) = V'_{j,1}\tau_w V'_{j,2}$ for each i, j, where

$$V_{i,1} = v_0 w v_1 w \dots v_{i-2} w v_{i-1}, \quad V_{i,2} = v_i w v_{i+1} \dots v_{n-1} w v_n,$$

$$V'_{j,1} = v'_0 w v'_1 w \dots v'_{j-2} w v'_{j-1}, \quad V'_{j,2} = v'_j w v'_{j+1} \dots v'_{n'-1} w v'_{n'}.$$

Clearly $|V_{1,1}| < |V_{2,1}| < \cdots < |V_{n,1}|$ and $|V'_{1,1}| < |V'_{2,1}| < \cdots < |V'_{n',1}|$ hold. By Claim A, it follows that

$$\tau_w = V_{i,1} w V_{i,2} = V'_{j,1} w V'_{j,2} (= \beta_w), \quad i = 1, \dots, n, \quad j = 1, \dots, n'.$$

By $(*)$, for each i there is an index j satisfying $r_i(\tau) = r_j(\beta)$. Hereafter we prove that $r_i(\tau) = r_j(\beta)$ implies $|V_{i,1}| = |V'_{j,1}|$. We suppose the converse, and $|V_{i,1}| < |V'_{j,1}|$, and so $|V_{i,2}| > |V'_{j,2}|$. Then since $V_{i,1}$ and $V'_{j,1}$ are prefixes of τ_w, $V'_{j,1} = V_{i,1} V$ hold for some nonempty string V. Similarly $V_{i,2} = V' V'_{j,2}$ holds for some nonmultiple string V'. By $\tau_w = \beta_w$, it follows that $\tau_w = V_{i,1} w (V' V'_{j,2}) = (V_{i,1} V) w V'_{j,2}$, and so

$$wV' = Vw.$$

By Lemma 25(ii), we have $w \preccurlyeq_p V^*$, $w \preccurlyeq_s V'^*$ and $V' = u''u'$, where $u' = \rho_p(w, V)$ and $V = u'u''$ for some $u'' \in \Sigma^+$.

On the other hand, by $r_i(\tau) = r_j(\beta)$ and $wV' = Vw$, we have

$$V_{i,1} V = V V_{i,1}, \quad V'_{j,2} V' = V' V'_{j,2}.$$

Applying Lemma 25(i), the above yield $V_{i,1}, V \in \{u\}^+$ and $V', V_{j,2} \in \{\hat{u}\}^+$ for some nonmultiple strings u and $\hat{u}$. By $u' = \rho_p(w, V)$ and $V \in \{u\}^+$, it follows that $w \preccurlyeq_p u^*$ and $\rho_p(w, u) = \rho_p(u', u)$ for the nonmultiple string u. Let $v' = \rho_p(u', u)$ and $u = v'v''$ for some nonempty string v''. Then clearly $\rho_s(u'', v''v') = v''$ holds. Since $V' = u''u'$ holds, it implies that $V' \in \{v''v'\}^+$. Remember that the string $\hat{u}$ is the nonmultiple string satisfying $V' \in \{\hat{u}\}^+$. Thus it leads that $v''v' \in \{\hat{u}\}^+$. By $\tau_w = (V_{i,1} w)(V' V'_{j,2})$, $\rho_s(w, v''v') = v'$, $\rho_s(V_{i,1} w, v''v') = v'$ and $V' V'_{j,2} \in \{\hat{u}\}^+$, we finally get $w \preccurlyeq_s \hat{u}$, $\tau_w \preccurlyeq_s \hat{u}$ and $\rho_s(w, \hat{u}) = \rho_s(\tau_w, \hat{u}) = \rho_s(v', \hat{u})$. It means that $w \preccurlyeq_p \hat{u}$, $\tau_w \preccurlyeq_p \hat{u}$ and $\rho_p(w, \hat{u}) = \rho_p(\tau_w, \hat{u})$. It contradicts to our assumption. Therefore, $r_i(\tau) = r_j(\beta)$ implies $|V_{i,1}| = |V'_{j,1}|$.

It implies together with $(*)$ that $n = n'$ and $V_{i,1} = V'_{i,1}$ for $i = 1, \dots, n$. Thus, $v_0 = \varepsilon$ (resp. $v_n = \varepsilon$) if and only if $v'_0 = \varepsilon$ (resp. $v'_n = \varepsilon$). As easily seen, $V_{i,1} = V'_{i,1}$ for each i implies $v_i = v'_i$ for every i. Hence we have $\tau = \beta$. It means that $L(w, \tau) = L(w, \beta)$, and a contradiction. □

5.2. The case a PFS $\Gamma = (\varepsilon, \tau)$, τ is nonmultiple

Here we consider a PFS $\Gamma = (\varepsilon, \tau)$, provided that τ_ε is a nonmultiple string. Let us put $\tau = u_0 X_1 u_1 \dots u_{k-1} X_k u_k$ for $u_0, u_k \in \Sigma^*$, $u_i \in \Sigma^+ (i = 1, \dots, k-1)$ and $X_i \in X^+$ for $i = 1, \dots, k$. Let $s = \tau_\varepsilon = u_0 u_1 \dots u_k$ and $l = |s|$.

For the pattern τ and $t \geqslant 1$, let us define a set $P_{\tau,t}$ of strings over $\Sigma \cup X \cup \{\langle, \rangle\}$ recursively as follows:

$$P_{\tau,1} = \bigcup_{D_\varepsilon \subsetneq var(\tau)} \{\tau\{x := \varepsilon \mid x \in D_\varepsilon\}\}$$

$$P_{\tau,t} = \bigcup_{\substack{D_\varepsilon, D_b \subseteq var(\tau) \\ D_\varepsilon \neq var(\tau) \\ D_b \cap D_\varepsilon = \phi}} \bigcup_{S \in U(D_b, P_{\tau,t-1})} \{\tau(\{x := \varepsilon \mid x \in D_\varepsilon\} \cup \{x := \langle \xi \rangle \mid (x, \xi) \in S\})\},$$

where

$$U(D_b, P_{\tau,t-1}) = \left\{ \{(x_1, \xi_1), \dots, (x_n, \xi_n)\} \subseteq D_b \times P_{\tau,t-1} \;\middle|\; \begin{array}{l} n = \sharp D_b, \\ \{x_1, \dots, x_n\} = D_b, \\ \{\xi_1, \dots, \xi_n\} \subseteq P_{\tau,t-1} \end{array} \right\}.$$

Then we put $P_\tau = \bigcup_{t=1}^{\infty} P_{\tau,t}$.

For each $\xi \in P_\tau$, the notation $I(\xi)$ means the constant string over Σ obtained from ξ by deleting all brackets $\langle\rangle$, and substituting $s (= \tau_\varepsilon)$ to each variable in ξ. The string ξ is called a *bracket expression* for a string $I(\xi)$. Let us put $I(P_\tau) = \{I(\xi) \mid \xi \in P_\tau\}$. Then the next result can be shown similarly to the proof of Lemma 4.

Lemma 29. *Let $\Gamma = (\varepsilon, \tau)$ be a PFS. Then we have*
(i) $I(P_\tau) = \Gamma_\tau - \{s\}$.
(ii) *For any $\xi \in P_\tau$ and $D \subseteq var(\xi)$, $I(\xi\{x := \varepsilon \text{ for } x \in D\})$ belongs to Γ_τ.*

We define the following strings:

$$\hat{r}_i(\tau) = \tau(\{X_i := s\} \cup \{X_m := \varepsilon \mid m \neq i\}),$$

$$t_{i,j}(\tau) = \tau(\{X_i := s^j\} \cup \{X_m := \varepsilon \mid m \neq i\}), \quad j = 1, \dots, |X_i|,$$

for $i = 1, \dots, k$, where $\{X_i := s\}$ substitutes s for the first variable of X_i and ε for the rest of variables, $\{X_i := s^j\}$ substitutes s for the first j variables of X_i and ε for the rest of variables, and $\{X_m := \varepsilon\}$ substitutes ε for every variable in X_i. Then clearly

$$\{\varepsilon, s\} \cup \{\hat{r}_i(\tau), t_{i,j}(\tau) \mid j = 1, \dots, |X_i|,\ i = 1, \dots, k\} \subseteq T(\Gamma).$$

Note that $t_{i,j}(\tau) = I(u_0 \dots u_{i-1} x_1 x_2 \dots x_j u_i \dots u_n)$ holds.

As easily seen, the empty string ε is the unique shortest string of $L(\Gamma)$ and $T(\Gamma)$, and s is the second shortest string of them. Moreover, the third shortest strings of them are $\hat{r}_i(\tau)$s whose lengths are $2l$ for $l = |s|$.

Lemma 30. *Let $\Gamma = (\varepsilon, \tau)$ be a PFS with a nonmultiple $s (= \tau_\varepsilon)$. Then the followings are valid*:
(i) *If $u_0 u_n \neq \varepsilon$, then $\hat{r}_i(\tau) \neq \hat{r}_j(\tau)$ for $i, j (1 \leqslant i < j \leqslant k)$.*
(ii) *If $u_0 u_n = \varepsilon$, then $\hat{r}_i(\tau) \neq \hat{r}_j(\tau)$ for $i, j (1 \leqslant i < j < k)$ and $\hat{r}_1(\tau) = \hat{r}_k(\tau)$.*
(iii) *There are no strings V_1 and V_2 such that $s^2 = V_1 s V_2$ and V_1, V_2 are not empty strings.*

Proof. (i) and (ii) can be shown similarly to that of Lemma 27. By Lemma 25(i), (iii) follows immediately. □

Theorem 31. *Let $\Gamma = (\varepsilon, \tau)$ be a reduced PFS, where τ_ε is a nonmultiple string. Then there does not exist a PFS $\Gamma' = (\alpha, \beta)$ such that*

$$T(\Gamma) \subseteq L(\Gamma') \subsetneq L(\Gamma).$$

Proof. We assume that there is a PFS $\Gamma' = (\alpha, \beta)$ satisfying the inclusion relations in our theorem. Then, since $\varepsilon \in L(\Gamma')$ and $\alpha \neq x$, clearly $\alpha = \varepsilon$ must hold. Let $\tau = u_0 X_1 u_1 \dots u_{k-1} X_k u_k$ and $\beta = u'_0 Y_1 \dots Y_{k'} u'_{k'}$ for some

$u_0, u'_0, u_k, u'_{k'} \in \Sigma^*, u_i, u'_j \in \Sigma^+ (i = 1, \ldots, k-1, j = 1, \ldots, k'-1)$ and $X_i, Y_j \in X^+$ for each i, j. Then since τ_ε and β_ε are the second shortest strings in $L(\Gamma)$ and $L(\Gamma')$, respectively, it follows that $s = \beta_\varepsilon$. Similarly we have

$$\{\hat{r}_i(\tau) \mid i = 1, \ldots, k\} = \{\hat{r}_j(\beta) \mid j = 1, \ldots, k'\}.$$

Claim A. *$k = k'$ and $u_i = u'_i$ for $i = 1, \ldots, k$.*

The proof of Claim A. Since s is nonmultiple, by Lemma 30(i), we have $\hat{r}_i(\tau) \neq \hat{r}_j(\tau)$ for $i \neq j$. By the above, it means that $k = k'$. Similar to the proof of Theorem 28, we can show that $u_i = u'_i$ for each i.

By the above, we have $\beta = u_0 Y_1 u_1 Y_2 \ldots Y_k u_k$.

Claim B. *$|X_i| \leqslant |Y_i|$ for $i = 1, \ldots, k$.*

The proof of Claim B. Since $t_{i,j}(\tau) \in \Gamma'_\beta$ for $j = 1, \ldots, |X_i|$, by Lemma 29(i), there is a bracket expression in P_β for each $t_{i,j}(\tau)$. We show by mathematical induction on j that $V_1 x_1 \ldots x_j V_2$ is the unique bracket expression for $V_1 s^j V_2$ within Γ', where $V_1 = u_0 \ldots u_{i-1}$ and $V_2 = u_i \ldots u_k$.

For the case of $j = 1$, clearly the string $V_1 x_1 V_2$ is the unique bracket expression for $V_1 s V_2$. Let $\xi_j \in P_\beta$ be a bracket expression for $t_{i,j}(\tau)$, and let ξ'_j be the bracket expression by substituting ε to the left most variable x_1 of ξ_j. By Lemma 30(iii), the place of the string s in $I(\xi_j)$ substituted to the variable x_1 must be the same one of either s of $s \cdot s \cdots s$ in $V_1 s^j V_2$. Thus, by Lemmas 29(ii) and 30(iii), ξ'_j is the bracket expression for $t_{i,j-1}$. By induction hypothesis, $\xi'_j = V_1 x_1 x_2 \ldots x_{j-1} V_2$. It means that $\xi_j = V_1 x_1 \ldots x_j V_2$. Since $t_{i,|X_i|}(\tau) \in \Gamma'_\beta$, we obtain $|Y_i| \geqslant |X_i|$.

Claims A and B imply that $\Gamma_\tau \subseteq \Gamma'_\beta$, and a contradiction. □

By Theorems 23, 24, 28 and Theorem 31, it follows that:

Theorem 32. *Let Γ be a PFS in $\mathcal{PFS}$. Then the set $T(\Gamma)$ is a finite tell tale set of the language $L(\Gamma)$.*

By Theorems 32 and 10, we obtain the following main theorem.

Theorem 33. *The class $\mathcal{PFSL}$ is inferable from positive examples.*

6. Conclusions

In this paper, we have investigated the learnability of the class $\mathcal{PFSL}$ of erasing PFS languages from positive examples. In language learning in Gold framework, *finite elasticity* is well-known as a powerful sufficient condition for learnability. In this paper, our class $\mathcal{PFSL}$, unfortunately, is shown not to have finite elasticity. Our approach to the learnability is to present concretely a finite tell tale set for each PFS language. The existence of such a finite subset for each language was shown by Angluin to be necessary for learnability. We have introduced the particular type of the finite subset $T(\Gamma)$ for each PFS Γ. In order to show that $T(\Gamma)$ is a finite tell tale set of $L(\Gamma)$, the inclusion problem for PFS languages is a very important key. We gave some solutions of the problem only for the case that patterns π in heads contain some variables. To solve the problem for the case of $\pi \in \Sigma^*$, it seems to be necessary to investigate equation problems on constant strings similar to Lemma 25. It is a future work to solve the inclusion problem for PFS languages for the other cases.

In this paper, we have shown that the class $\mathcal{PFSL}$ is learnable from positive examples. A target PFS $\Gamma = (\pi, \tau)$ is assumed $\tau_\pi \not\leqslant_p u^*$ or $\rho_p(\tau_\pi, u) \neq \pi$ for any nonmultiple string u with $|u| < |\tau_\pi|$ if $\pi \in \Sigma^*$. It is a future work to find a finite tell tale set of such a PFS language $L(\Gamma)$.

References

[2] D. Angluin, Inductive inference of formal languages from positive data, Inform. and Control 45 (1980) 117–135.
[3] H. Arimura, T. Shinohara, S. Otsuki, Finding Minimal Generalizations for Unions of Pattern Languages and its Application to Inductive Inference from Positive Data, Lecture Notes in Computer Science, Vol. 775, Springer, Berlin, 1994, pp. 646–660.
[4] S. Arikawa, T. Shinohara, A. Yamamoto, Learning elementary formal system, Theoret. Comput. Sci. 95 (1992) 97–113.

[5] E.M. Gold, Language identification in the limit, Inform. and Control 10 (1967) 447–474.
[6] T. Moriyama, M. Sato, Properties of language classes with finite elasticity, IEICE Trans. Inform. Syst. E78-D (5) (1995) 532–538.
[7] T. Motoki, T. Shinohara, K. Wright, The correct definition of finite elasticity: corrigendum to identification of unions, in: Proc. Fourth Annu. Workshop on Computational Learning Theory, 1991, pp. 375–375.
[8] Y. Mukouchi, S. Arikawa, Towards a mathematical theory of machine discovery from facts, Theoret. Comput. Sci. 137 (1995) 53–84.
[9] Y. Mukouchi, M. Sato, Learning Languages Generated by Elementary Formal Systems and its Application to SH Languages, Lecture Notes in Artificial Intelligence, Springer, Berlin, in preparation.
[10] D. Reidenbach, Result on Inductive Inference of Extended Pattern Languages, Lecture Notes in Artificial Intelligence, Vol. 2533, Springer, Berlin, 2002, pp. 308–320.
[11] D. Reidenbach, On the Learnability of E-pattern Languages over Small Alphabets, Lecture Notes in Computer Science, Vol. 3120, Springer, Berlin, 2004, pp. 140–154.
[12] M. Sato, Inductive inference of formal languages, Bull. Inform. Cybernet. 27 (1) (1995) 85–106.
[13] T. Shinohara, Polynomial Time Inference of Extended Regular Pattern Languages, Lecture Notes in Computer Science, Vol. 147, Springer, Berlin, 1982, pp. 115–127.
[14] T. Shinohara, Inductive inference of formal systems from positive data, Bull. Inform. Cybernet. 22 (1986) 9–18.
[15] T. Shinohara, Inductive inference from positive data is powerful, in: Proc. Third Annu. Workshop on Computational Learning Theory, 1990, pp. 97–110.
[16] T. Shinohara, Inductive inference of monotonic formal systems from positive data, New Generation Comput. 8 (1991) 371–384.
[17] T. Shinohara, Rich classes inferable from positive data, Inform. and Comput. 108 (1994) 175–186.
[18] R.M. Smullyan, Theory of Formal Systems, Princeton University Press, Princeton, NJ, 1961.
[19] J. Uemura, M. Sato, Compactness and Learning of Unions of Erasing Regular Pattern Languages, Lecture Notes in Artificial Intelligence, Vol. 2533, Springer, Berlin, 2002, pp. 293–307.
[20] K. Wright, Identification of unions of languages drawn from positive data, in: Proc. Second Annu. Workshop on Computational Learning Theory, 1989, pp. 328–333.

Further reading

[1] D. Angluin, Finding patterns common to a set of strings, in: Proc. 11th Annu. Symp. on Theory of Computing, 1979, pp. 130–141.